IILM UNIVERSITY Greater Noida

Lab File
Of
Programming in C
(UCS1003)

(2025 - 2026)

Submitted To:

Submitted by:

Name:
Roll No.:
Section:

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S.no	Name of the practical	Grade	Date	SIGNATURE

1. Write a Program to print different data types in ‘C’ and their ranges

<u>Program:</u>

```
#include <stdio.h>
#include <limits.h>
#include <float.h>

int main() {
    printf("----------- Integer Data Types------------ \n");
    printf("char: %zu bytes | Range: %d to %d\n", sizeof(char), CHAR_MIN, CHAR_MAX);
    printf("unsigned char: %zu bytes | Range: 0 to %u\n", sizeof(unsigned char), UCHAR_MAX);
    printf("short: %zu bytes | Range: %d to %d\n", sizeof(short), SHRT_MIN, SHRT_MAX);
    printf("unsigned short: %zu bytes | Range: 0 to %u\n", sizeof(unsigned short), USHRT_MAX);
    printf("int: %zu bytes | Range: %d to %d\n", sizeof(int), INT_MIN, INT_MAX);
    printf("unsigned int: %zu bytes | Range: 0 to %u\n", sizeof(unsigned int), UINT_MAX);
    printf("long: %zu bytes | Range: %ld to %ld\n", sizeof(long), LONG_MIN, LONG_MAX);
    printf("unsigned long: %zu bytes | Range: 0 to %lu\n", sizeof(unsigned long), ULONG_MAX);
    printf("long long: %zu bytes | Range: %lld to %lld\n", sizeof(long long), LLONG_MIN, LLONG_MAX);
    printf("unsigned long long: %zu bytes | Range: 0 to %llu\n", sizeof(unsigned long long),
ULLONG_MAX);

    printf("\n----------- Floating-Point Data Types ----------- \n");
    printf("float: %zu bytes | Range: %e to %e\n", sizeof(float), FLT_MIN, FLT_MAX);
    printf("double: %zu bytes | Range: %e to %e\n", sizeof(double), DBL_MIN, DBL_MAX);
    printf("long double: %zu bytes | Range: %Le to %Le\n", sizeof(long double), LDBL_MIN,
LDBL_MAX);

    return 0;
}
```

<u>Output:</u>

```
----------- Integer Data Types -----------
char: 1 bytes | Range: -128 to 127
unsigned char: 1 bytes | Range: 0 to 255
short: 2 bytes | Range: -32768 to 32767
unsigned short: 2 bytes | Range: 0 to 65535
int: 4 bytes | Range: -2147483648 to 2147483647
unsigned int: 4 bytes | Range: 0 to 4294967295
long: 8 bytes | Range: -9223372036854775808 to 9223372036854775807
unsigned long: 8 bytes | Range: 0 to 18446744073709551615
long long: 8 bytes | Range: -9223372036854775808 to 9223372036854775807
unsigned long long: 8 bytes | Range: 0 to 18446744073709551615

----------- Floating-Point Data Types -----------
float: 4 bytes | Range: 1.175494e-38 to 3.402823e+38
double: 8 bytes | Range: 2.225074e-308 to 1.797693e+308
long double: 16 bytes | Range: 3.362103e-4932 to 1.189731e+4932
```

2. Write a C program to find the sum and average of three numbers.

Program:

```
#include <stdio.h>

int main() {
    int num1, num2, num3;
    float sum, average;

    // Input three numbers
    printf("Enter three numbers: ");
    scanf("%d %d %d", &num1, &num2, &num3);

    // Calculate sum and average
    sum = num1 + num2 + num3;
    average = sum / 3;

    // Display results
    printf("Sum = %.2f\n", sum);
    printf("Average = %.2f\n", average);

    return 0;
}
```

Output:

```
Enter three numbers: 10 20 30
Sum = 60.00
Average = 20.00
```

3. Write a program to swap the values of two variables with and without using the third variable.

Program:

```
#include <stdio.h>

int main() {
   int a, b, temp;

   // Input values
   printf("Enter two numbers: ");
   scanf("%d %d", &a, &b);

   // Display original values
   printf("\nBefore swapping:\n");
   printf("a = %d, b = %d\n", a, b);

   // ---------- Method 1: Using a third variable ----------
   temp = a;
   a = b;
   b = temp;

   printf("\nAfter swapping (using third variable):\n");
   printf("a = %d, b = %d\n", a, b);

   // ---------- Method 2: Without using a third variable ----------
   a = a + b;  // sum of both
   b = a - b;  // get original a
   a = a - b;  // get original b

   printf("\nAfter swapping (without using third variable):\n");
   printf("a = %d, b = %d\n", a, b);

   return 0;
}
```

Output:

```
Enter two numbers: 10 20

Before swapping:
a = 10, b = 20

After swapping (using third variable):
a = 20, b = 10

After swapping (without using third variable):
a = 10, b = 20
```

4. Write a Program to read radius value from the keyboard and calculate the area of circle and print the result in both floating and exponential notation.

C print.c > ...

```c
#include <stdio.h>
int main() {
    float PI=3.141592; // pi value upto 6th decimal
    double radius, area;

    // Read radius from keyboard
    printf("Enter the radius of the circle: ");
    scanf("%lf", &radius);

    // Calculate area of circle
    area = PI * radius * radius;

    // Print results
    printf("\nArea of the circle:\n");
    printf("Floating-point notation: %.6f\n", area);
    printf("Exponential notation: %.6e\n", area);
    return 0;
}
/*
%.6f displays floating-point notation with 6 decimal places
%.6e displays exponential notation with 6 decimal places
*/
```

Output:-

```
Area of the circle:
Floating-point notation: 95.033159
Exponential notation: 9.503316e+001
```

5. Write a program to find whether the given year is a leap year or not.
A year is a leap year if it satisfies either of these conditions:

1. Divisible by 4 BUT not divisible by 100
    - Example: 2024, 2020, 2016

2. Divisible by 400
    - Example: 2000, 2400

- Divisible by 4: Most leap years are every 4 years
- But not by 100: Century years (1700, 1800, 1900) are NOT leap years
- Except by 400: Every 400 years, we have a leap year to correct the calendar

leap_year.c > main()

```c
#include <stdio.h>

int main() {
    int year;
    printf("Enter a year: ");
    scanf("%d", &year);

    if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) {
        printf("%d is a LEAP YEAR\n", year);
    } else {
        printf("%d is NOT a LEAP YEAR\n", year);
    }
    return 0;
}
```

```
Enter a year: 2024
2024 is a LEAP YEAR
```

6 Write a program that checks whether the two numbers entered by the user are equal or not

same_num.c > ...

```c
#include <stdio.h>
int main() {
    int num1, num2;
    // Read two numbers from user
    printf("Enter first number: ");
    scanf("%d", &num1);
    printf("Enter second number: ");
    scanf("%d", &num2);
    // Check if the numbers are equal
    if (num1 == num2) {
        printf("The numbers %d and %d are EQUAL\n", num1, num2);
    } else {
        printf("The numbers %d and %d are NOT EQUAL\n", num1, num2);
    }
    return 0;
}
```

```
Enter first number: 10
Enter second number: 10
The numbers 10 and 10 are EQUAL
```

7.Write a program to find the greatest of three numbers using nested if else?

gratest.c > ...

```c
#include <stdio.h>
int main() {
    int a, b, c;
    printf("Enter three numbers: ");
    scanf("%d %d %d", &a, &b, &c);
    printf("You entered: %d, %d, %d\n", a, b, c);
    printf("The greatest number is: ");
    if (a > b) {
        if (a > c) {
            printf(" A is gratest %d\n", a);
        } else {
            printf(" C is gratest %d\n", c);
        }
    } else {
        if (b > c) {
            printf("B is gratest %d\n", b);
        } else {
            printf("C is gratest %d\n", c);
        }
    }
    return 0;
}
```

```
Enter three numbers: 10 20 30
You entered: 10, 20, 30
The greatest number is: C is gratest 30
```

8 Write a program that finds whether a given number is even or odd.

```c
// even_odd.c > main()
#include <stdio.h>

int main() {
    int number;
    printf("Enter a number: ");
    scanf("%d", &number);
    if (number % 2 == 0) { //condition checking
        printf("%d is an even number.\n", number);
    } else {
        printf("%d is an odd number.\n", number);
    }
    return 0;
}
```

```
Enter a number: 10
10 is an even number.
PS C:\Users\Animesh\Desktop\c_programming>
```

Tarnary operator

tarnary_even_odd.c > main()

```c
#include <stdio.h>
int main() {
    int number;

    printf("Enter a number: ");
    scanf("%d", &number);
    // Using ternary operator
    (number % 2 == 0) ? printf("%d is an even number.\n", number)
                      : printf("%d is an odd number.\n", number);

    return 0;
}
```


Question no 5 and 9 same pattern

10. write a program that accepts marks of five subjects and finds percentage and prints grades according to

the following criteria: Between 90-100%----- Print 'A'

80-90% --------------------- Print 'B' 60-80% --------------------- Print 'C'
Below 60% ------------ Print 'D' in code

subject.c > main()

```c
#include <stdio.h>
int main() {
    float subject1, subject2, subject3, subject4, subject5;
    float total, percentage;

    // Input marks for five subjects
    printf("Enter marks for subject 1: ");
    scanf("%f", &subject1);
    printf("Enter marks for subject 2: ");
    scanf("%f", &subject2);

    printf("Enter marks for subject 3: ");
    scanf("%f", &subject3);

    printf("Enter marks for subject 4: ");
    scanf("%f", &subject4);

    printf("Enter marks for subject 5: ");
    scanf("%f", &subject5);

    // Calculate total and percentage
    total = subject1 + subject2 + subject3 + subject4 + subject5;
    percentage = (total / 500) * 100;

    printf("\n--- Results ---\n");
    printf("Total Marks: %.2f/500\n", total);
    printf("Percentage: %.2f%%\n", percentage);

    // Determine grade
    if (percentage >= 90 && percentage <= 100) {
        printf("Grade: A\n");
    } else if (percentage >= 80 && percentage < 90) {
        printf("Grade: B\n");
    } else if (percentage >= 60 && percentage < 80) {
        printf("Grade: C\n");
    } else if (percentage < 60) {
        printf("Grade: D\n");
    } else {
        printf("Invalid percentage!\n");
    }
    return 0;
}
```

```
Enter marks for subject 1: 50
Enter marks for subject 2: 45
Enter marks for subject 3: 65
Enter marks for subject 4: 55
Enter marks for subject 5: 45

--- Results ---
Total Marks: 260.00/500
Percentage: 52.00%
Grade: D
PS C:\Users\Animesh\Desktop\c_programming>
```

11. write a program that takes two operands and one operator from the user, perform the operation, and prints the result by using Switch statement. in c.

C calc.c > main()

```c
int main() {
    double num1, num2, result;
    char operator;

    // Input from user
    printf("Enter first operand: ");
    scanf("%lf", &num1);

    printf("Enter operator (+, -, *, /): ");
    scanf(" %c", &operator);

    printf("Enter second operand: ");
    scanf("%lf", &num2);

    // Perform operation using switch statement
    switch (operator) {
        case '+':
            result = num1 + num2;
            printf("%.2lf + %.2lf = %.2lf\n", num1, num2, result);
            break;

        case '-':
            result = num1 - num2;
            printf("%.2lf - %.2lf = %.2lf\n", num1, num2, result);
            break;

        case '*':
            result = num1 * num2;
            printf("%.2lf * %.2lf = %.2lf\n", num1, num2, result);
            break;

        case '/':
            if (num2 != 0) {
                result = num1 / num2;
                printf("%.2lf / %.2lf = %.2lf\n", num1, num2, result);
            } else {
                printf("Error: Division by zero is not allowed!\n");
            }
            break;
        default:
            printf("Error: Invalid operator! Please use +, -, *, or /\n");
            break;
    }
    return 0;
}
```

```
PS C:\Users\Animesh\Desktop\c_programming>
Enter first operand: 20
Enter operator (+, -, *, /): +
Enter second operand: 30
20.00 + 30.00 = 50.00
PS C:\Users\Animesh\Desktop\c_programming>
```

12. Write a program to print the sum of all numbers up to a given number

sum.c > main()

```c
#include <stdio.h>
int main() {
    int n, sum = 0;

    // user input
    printf("Enter a positive integer: ");
    scanf("%d", &n);

    // Calculate sum using  forloop
    for(int i = 1; i <= n; i++) {
        sum += i;
    }
    printf("Sum of numbers from 1 to %d is: %d\n", n, sum);
    return 0;
}
```

```
Enter a positive integer: 10
Sum of numbers from 1 to 10 is: 55
PS C:\Users\Animesh\Desktop\c_programming>
```

13 Write a program to find the factorial of a given number.

C fact.c > main()

```c
#include <stdio.h>

int main() {
    int n, i;
    float factorial = 1;

    printf("Enter a positive integer: ");
    scanf("%d", &n);
    // Calculate factorial
    for(i = 1; i <= n; i++) {
        factorial *= i;
    }
    printf("Factorial of %d = %.2f\n", n, factorial);

    return 0;
}
```


Using while loop.

```c
#include <stdio.h>
int main() {
    int n;
    float factorial = 1;
    printf("Enter a number: ");
    scanf("%d", &n);
    int i = 1;
    while(i <= n) {
        factorial *= i;
        i++;
    }
    printf("Factorial = %.2f\n", factorial);
    return 0;
}
```

14.Write a program to print sum of even and odd numbers from 1 to N numbers

C sum_even_odd.c > main()

```c
#include <stdio.h>
int main() {
    int n, i;
    int even_sum = 0, odd_sum = 0;

    printf("Enter a number N: ");
    scanf("%d", &n);

    for(i = 1; i <= n; i++) {
        if(i % 2 == 0) {
            even_sum += i;  // Add to even sum
        } else {
            odd_sum += i;   // Add to odd sum
        }
    }
    printf("Sum of even numbers from 1 to %d: %d\n", n, even_sum);
    printf("Sum of odd numbers from 1 to %d: %d\n", n, odd_sum);
    return 0;
}
```

```
Enter a number N: 20
Sum of even numbers from 1 to 20: 110
Sum of odd numbers from 1 to 20: 100
PS C:\Users\Animesh\Desktop\c_programming>
```

15 Write a program to print the Fibonacci series.

Ans :- The Fibonacci series is a sequence where each number is the sum of the two preceding ones:

0, 1, 1, 2, 3, 5, 8, 13, 21, 34

- First two terms are always 0 and 1
- Each subsequent term = sum of previous two terms
- Method 1 handles first two terms inside the loop
- Method 2 prints first two terms separately
- Method 3 stops when the next term exceeds a maximum value

```c
#include <stdio.h>
int main() {
    int n, i;
    int first = 0, second = 1, next;
    printf("Enter the number of terms: ");
    scanf("%d", &n);
    printf("Fibonacci Series: ");
    for(i = 0; i < n; i++) {
        if(i <= 1) {
            next = i;  // First two terms are 0 and 1
        } else {
            next = first + second;
            first = second;
            second = next;
        }
        printf("%d ", next);
    }
    return 0;
}
```

```
Enter the number of terms: 5
Fibonacci Series: 0 1 1 2 3
PS C:\Users\Animesh\Desktop\c_programming>
```

16. Write a program to check whether the entered number is prime or not using function

What is a Prime Number?

- A prime number is a natural number greater than 1
- It has exactly two distinct factors: 1 and itself
- Examples: 2, 3, 5, 7, 11, 13, 17, 19, 23...

C prime_no.c > main()

```c
#include <stdio.h>
int main()
{
    int n, i, flag = 0;
    printf("Enter a number: ");
    scanf("%d", &n);
    if (n <= 1) {
        printf("%d is not prime\n", n);
        return 0;
    }
    for(i = 2; i <= n/2; i++) {
        if(n % i == 0) {
            flag = 1;
            break;
        }
    }
    if(flag == 0)
        printf("%d is prime\n", n);
    else
        printf("%d is not prime\n", n);
    return 0;
}
```

```
Enter a number: 5
5 is prime
PS C:\Users\Animesh\Desktop\c_programming>
```

17. Write a program to find the reverse of a number using a function.

```c
#include <stdio.h>
int reverseNumber(int num); // function declaration
int main() {
    int number, reversed;

    printf("Enter a number: ");
    scanf("%d", &number);

    reversed = reverseNumber(number);

    printf("Original number: %d\n", number);
    printf("Reversed number: %d\n", reversed);

    return 0;
}
// Function to reverse a number
int reverseNumber(int num) {
    int reversed = 0;
    int remainder;
    // Reverse the number
    while (num != 0) {
        remainder = num % 10;
        reversed = reversed * 10 + remainder;
        num = num / 10;
    }
    return reversed;
}
```

```
Enter a number: 12345
Original number: 12345
Reversed number: 54321
PS C:\Users\Animesh\Desktop\c_
```

18. Write a program to print Armstrong numbers from 1 to 100 using a function.

An Armstrong number (also called narcissistic number) is a number that equals the sum of its own digits each raised to the power of the number of digits.

Example: $153 = 1^3 + 5^3 + 3^3 = 1 + 125 + 27 = 153$

C armstrong_no.c > main()

```c
#include<stdio.h>
#include<conio.h>
int main()
{
    int number,temp,sum,remainder;
    printf("The Armstrong number between 1-100 \n");
    for(number=1;number<=1000;number++)
    {
        sum=0;
        temp=number;
        while(temp!=0)
        {
            remainder=temp%10;
            sum=sum+(remainder*remainder*remainder);
            temp=temp/10;
        }
        if(sum==number)
        printf("%d \n",number);
    }
    return 0;
}
```

```
The Armstrong number between 1-100
1
153
370
371
407
PS C:\Users\Animesh\Desktop\c_programming>
```

Using function

```c
arms1.c > isArmstrong(int)
#include<stdio.h>
int isArmstrong(int number);
int main() {
    int number;

    printf("The Armstrong numbers between 1-1000 are: \n");

    for(number = 1; number <= 1000; number++) {
        if(isArmstrong(number)) {
            printf("%d \n", number);
        }
    }
    return 0;
}
// Function to check if a number is Armstrong
int isArmstrong(int number) {
    int temp, sum = 0, remainder;

    temp = number;
    while(temp != 0) {
        remainder = temp % 10;
        sum = sum + (remainder * remainder * remainder);
        temp = temp / 10;
    }
    // Return 1 if Armstrong, 0 if not
    if(sum == number)
        return 1;
    else
        return 0;
}
```

19. Write a program to find the factorial of number using recursive

recv_fact.c > fact(int)

```c
#include <stdio.h>
int fact(int n);
int main() {
    int num = 5;
    printf("Factorial of %d = %d", num, fact(num));
    return 0;
}
// Very simple recursive factorial
int fact(int n) {
    if (n==0) return 1;
    return n * fact(n - 1);
}
```

20 Write a program that simply takes elements of the array from the user and finds the sum of these elements using function.

```c
#include <stdio.h>
void inputArray(int arr[], int n);
int arraySum(int arr[], int n);
int main() {
    int n;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    int arr[n];
    inputArray(arr, n);
    int sum = arraySum(arr, n);
    printf("Sum = %d\n", sum);
    return 0;
}

void inputArray(int arr[], int n) {
    printf("Enter %d elements:\n", n);
    for(int i = 0; i < n; i++) {
        printf("Element %d: ", i + 1);
        scanf("%d", &arr[i]);
    }
}
int arraySum(int arr[], int n) {
    int total = 0;
    for(int i = 0; i < n; i++) {
        total += arr[i];
    }
    return total;
}
```

```
Enter number of elements: 5
Enter 5 elements:
Element 1: 10
Element 2: 20
Element 3: 30
Element 4: 40
Element 5: 50
Sum = 150
```

21. Write a program that inputs two arrays and saves the sum of corresponding elements of these arrays in a third array and prints them.

First part

```c
#include <stdio.h>
void inputArray(int arr[], int size, char arrayName);
void sumArrays(int arr1[], int arr2[], int sum[], int size) ;
void displayArray(int arr[], int size, char arrayName);
int main() {
    int size;

    printf("Enter the size of arrays: ");
    scanf("%d", &size);

    int arr1[size], arr2[size], result[size];

    // Input both arrays
    inputArray(arr1, size, 'A');
    inputArray(arr2, size, 'B');

    // Calculate sum
    sumArrays(arr1, arr2, result, size);

    // Display all arrays
    printf("\n");
    displayArray(arr1, size, 'A');
    displayArray(arr2, size, 'B');
    displayArray(result, size, 'C');

    return 0;
}
```

2nd part

```c
// Function to insert element inside the array
void inputArray(int arr[], int size, char arrayName) {
    printf("Enter %d elements for array %c:\n", size, arrayName);
    for(int i = 0; i < size; i++) {
        printf("Element %d: ", i + 1);
        scanf("%d", &arr[i]);
    }
}
// Function to calculate sum of two arrays
void sumArrays(int arr1[], int arr2[], int sum[], int size) {
    for(int i = 0; i < size; i++) {
        sum[i] = arr1[i] + arr2[i];
    }
}
// Function to display array
void displayArray(int arr[], int size, char arrayName) {
    printf("Array %c: ", arrayName);
    for(int i = 0; i < size; i++) {
        printf("%d ", arr[i]);
    }
    printf("\n");
}
```

```
Enter 3 elements for array A:
Element 1: 10
Element 2: 20
Element 3: 30
Enter 3 elements for array B:
Element 1: 20
Element 2: 30
Element 3: 40

Array A: 10 20 30
Array B: 20 30 40
Array C: 30 50 70
PS C:\Users\Animesh\Desktop\c_programming> cd "
```

22. Write a program to find the minimum and maximum element in the array.

min_max.c > main()

```c
#include <stdio.h>
int main() {
    int n, i;
    // Input the size of array
    printf("Enter the number of elements: ");
    scanf("%d", &n);
    int arr[n];

    // Input array elements
    printf("Enter %d elements:\n", n);
    for(i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
    // Initialize min and max with first element
    int min = arr[0];
    int max = arr[0];
    // Find min and max
    for(i = 1; i < n; i++) {
        if(arr[i] < min) {
            min = arr[i];
        }
        if(arr[i] > max) {
            max = arr[i];
        }
    }
    // Display results
    printf("Minimum element: %d\n", min);
    printf("Maximum element: %d\n", max);
    return 0;
}
```

```
Enter the number of elements: 5
Enter 5 elements:
10
11
7
8
16
Minimum element: 7
Maximum element: 16
PS C:\Users\Animesh\Desktop\c_programming>
```

23. Write a program to search an element in an array using Linear Search.

```c
#include <stdio.h>
int main() {
    int n, i, search, found = 0;
    // Input the size of array
    printf("Enter the number of elements: ");
    scanf("%d", &n);
    int arr[n];
    // Input array elements
    printf("Enter %d elements:\n", n);
    for(i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
    // Input element to search
    printf("Enter the element to search: ");
    scanf("%d", &search);
    // Linear Search
    for(i = 0; i < n; i++) {
        if(arr[i] == search) {
            printf("Element %d found at position %d\n", search, i + 1);
            found = 1;
            break;
        }
    }
    if(!found) {
        printf("Element %d not found in the array\n", search);
    }
    return 0;
}
```

```
Enter the number of elements: 5
Enter 5 elements:
10
20
30
40
50
Enter the element to search: 20
Element 20 found at position 2
PS C:\Users\Animesh\Desktop\c_programming>
```

24. Write a program to sort the elements of the array in ascending order using Bubble Sort technique.

```c
#include <stdio.h>
// Function to input array elements
void inputArray(int arr[], int n) {
    printf("Enter %d elements:\n", n);
    for(int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
}
// Function to display array elements
void displayArray(int arr[], int n) {
    for(int i = 0; i < n; i++) {
        printf("%d ", arr[i]);
    }
    printf("\n");
}
```

2nd part

```c
// Function to perform Bubble Sort
void bubbleSort(int arr[], int n) {
    int temp;
    for(int i = 0; i < n - 1; i++) {
        for(int j = 0; j < n - i - 1; j++) {
            if(arr[j] > arr[j + 1]) {
                // Swap elements
                temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }
}
```


3rd part

```c
int main() {
    int n;
    // Input the size of array
    printf("Enter the number of elements: ");
    scanf("%d", &n);
    int arr[n];
    // Input array using function
    inputArray(arr, n);
    // Display original array
    printf("\nOriginal array: ");
    displayArray(arr, n);
    // Sort array using Bubble Sort function
    bubbleSort(arr, n);
    // Display sorted array
    printf("Sorted array (Bubble Sort): ");
    displayArray(arr, n);
    return 0;
}
```

```
Enter the number of elements: 4
Enter 4 elements:
20
21
23
18

Original array: 20 21 23 18
Sorted array (Bubble Sort): 18 20 21 23
PS C:\Users\Animesh\Desktop\c_programming>
```

25.1 Write a code for counting the string length

```c
#include <stdio.h>
int string_length(char str[]);
int main() {
    char str[100];
    printf("Enter a string: ");
    scanf("%s", str);
    int len = string_length(str);
    printf("Length of '%s' is: %d\n", str, len);
    return 0;
}
int string_length(char str[]) {
    int length = 0;
    // Count characters until null terminator is found
    while(str[length] != '\0') {
        length++;
    }
    return length;
}
```

```
Enter a string: hello
Length of 'hello' is: 5
PS C:\Users\Animesh\Desktop\c_programming>
```

25.2 Write a code to copy a string into another empty string using function

C copy1.c > copy_string(char [], char [])

```c
#include <stdio.h>
// Simple string copy function
void copy_string(char dest[], char src[]) {
    int i = 0;
    // Copy each character until null terminator
    while(src[i] != '\0') {
        dest[i] = src[i];
        i++;
    }
    dest[i] = '\0'; // Add null terminator
}

int main() {
    char str1[100], str2[100];
    printf("Enter a string: ");
    scanf("%s", str1);

    copy_string(str2, str1);
    printf("Original string: %s\n", str1);
    printf("Copied string: %s\n", str2);
    return 0;
}
```

26 Write a program that finds the sum of diagonal elements of a m x n matrix.

Program:

```
#include <stdio.h>

int main() {
    int m, n;
    int matrix[10][10];  // you can increase size if needed
    int i, j, sum = 0;

    // Input matrix dimensions
    printf("Enter number of rows (m): ");
    scanf("%d", &m);
    printf("Enter number of columns (n): ");
    scanf("%d", &n);

    // Input matrix elements
    printf("\nEnter elements of the matrix:\n");
    for (i = 0; i < m; i++) {
        for (j = 0; j < n; j++) {
            printf("Element [%d][%d]: ", i + 1, j + 1);
            scanf("%d", &matrix[i][j]);
        }
    }

    // Display the matrix
    printf("\nThe matrix is:\n");
    for (i = 0; i < m; i++) {
        for (j = 0; j < n; j++) {
            printf("%d\t", matrix[i][j]);
        }
        printf("\n");
    }

    // Calculate sum of diagonal elements
    for (i = 0; i < m; i++) {
        for (j = 0; j < n; j++) {
            if (i == j)  // condition for main diagonal
                sum += matrix[i][j];
        }
    }

    // Display the result
    printf("\nSum of diagonal elements = %d\n", sum);

    return 0;
}
```

Output:

```
Enter number of rows (m): 3
Enter number of columns (n): 3

Enter elements of the matrix:
Element [1][1]: 1
Element [1][2]: 2
Element [1][3]: 3
Element [2][1]: 4
Element [2][2]: 5
Element [2][3]: 6
Element [3][1]: 7
Element [3][2]: 8
Element [3][3]: 9

The matrix is:
1   2   3
4   5   6
7   8   9

Sum of diagonal elements = 15
```

27 Write a program to implement strlen (), strcat (),strcpy () using the concept of Functions.

Program:

```
#include <stdio.h>

// Function to find the length of a string
int my_strlen(char str[]) {
    int i = 0;
    while (str[i] != '\0') {
        i++;
    }
    return i;
}

// Function to copy one string into another
void my_strcpy(char dest[], char src[]) {
    int i = 0;
    while (src[i] != '\0') {
        dest[i] = src[i];
        i++;
    }
    dest[i] = '\0';  // add null terminator
}

// Function to concatenate two strings
void my_strcat(char dest[], char src[]) {
    int i = 0, j = 0;
```

```
    // move to end of dest string
    while (dest[i] != '\0') {
        i++;
    }

    // copy src to the end of dest
    while (src[j] != '\0') {
        dest[i] = src[j];
        i++;
        j++;
    }
    dest[i] = '\0';  // add null terminator
}

int main() {
    char str1[100], str2[100], result[200];

    // Input strings
    printf("Enter first string: ");
    gets(str1);  // for simplicity (unsafe but okay for basic demo)
    printf("Enter second string: ");
    gets(str2);

    // strlen()
    printf("\nLength of first string = %d", my_strlen(str1));
    printf("\nLength of second string = %d", my_strlen(str2));

    // strcpy()
    my_strcpy(result, str1);
    printf("\n\nAfter copying str1 to result: %s", result);

    // strcat()
    my_strcat(result, str2);
    printf("\nAfter concatenation of str1 and str2: %s\n", result);

    return 0;
}
```

Output:

```
Enter first string: Hello
Enter second string: World

Length of first string = 5
Length of second string = 5

After copying str1 to result: Hello
After concatenation of str1 and str2: HelloWorld
```

28 Write a program Define a structure data type TRAIN_INFO. The type contains Train No.: integer type Train name: string Departure Time: aggregate type TIME Arrival Time: aggregate type TIME Start station: string

End station: string The structure type Time contains two integer members: hour and minute. Maintain a train timetable and implement the following operations: a. List all the trains (sorted according to train number) that depart from a particular section. b. List all the trains that depart from a particular station at a particular time. c. List all the trains that depart from a particular station within the next one hour of a given time. d. List all the trains between a pair of start station and end station

Program:

```
#include <stdio.h>
#include <string.h>

// Define TIME structure
struct TIME {
   int hour;
   int minute;
};

// Define TRAIN_INFO structure
struct TRAIN_INFO {
   int train_no;
   char train_name[50];
   struct TIME departure;
   struct TIME arrival;
   char start_station[50];
   char end_station[50];
};

// Function to compare times (in minutes)
int timeToMinutes(struct TIME t) {
   return t.hour * 60 + t.minute;
}

// Function to list all trains sorted by train number that depart from a given
station
void listTrainsFromStation(struct TRAIN_INFO trains[], int n, char station[])
{
   int i, j;
   struct TRAIN_INFO temp;

   // Sort trains by train number
   for (i = 0; i < n - 1; i++) {
      for (j = i + 1; j < n; j++) {
         if (trains[i].train_no > trains[j].train_no) {
            temp = trains[i];
            trains[i] = trains[j];
            trains[j] = temp;
         }
      }
```

```
    }

    printf("\nTrains departing from station '%s':\n", station);
    int found = 0;
    for (i = 0; i < n; i++) {
        if (strcmp(trains[i].start_station, station) == 0) {
            printf("%d - %s (%02d:%02d)\n", trains[i].train_no,
trains[i].train_name,
                   trains[i].departure.hour, trains[i].departure.minute);
            found = 1;
        }
    }
    if (!found)
        printf("No trains found from this station.\n");
}

// Function to list trains that depart from a station at a particular time
void listTrainsAtTime(struct TRAIN_INFO trains[], int n, char station[],
struct TIME time) {
    int i, found = 0;
    printf("\nTrains departing from '%s' at %02d:%02d:\n", station, time.hour,
time.minute);
    for (i = 0; i < n; i++) {
        if (strcmp(trains[i].start_station, station) == 0 &&
            trains[i].departure.hour == time.hour &&
            trains[i].departure.minute == time.minute) {
            printf("%d - %s\n", trains[i].train_no, trains[i].train_name);
            found = 1;
        }
    }
    if (!found)
        printf("No trains found at that time.\n");
}

// Function to list trains that depart from a station within next 1 hour of a
given time
void listTrainsWithinOneHour(struct TRAIN_INFO trains[], int n, char
station[], struct TIME time) {
    int i, found = 0;
    int givenTime = timeToMinutes(time);
    printf("\nTrains departing from '%s' within next 1 hour of %02d:%02d:\n",
           station, time.hour, time.minute);
    for (i = 0; i < n; i++) {
        if (strcmp(trains[i].start_station, station) == 0) {
            int depTime = timeToMinutes(trains[i].departure);
            if (depTime >= givenTime && depTime <= givenTime + 60) {
                printf("%d - %s (%02d:%02d)\n", trains[i].train_no,
trains[i].train_name,
                       trains[i].departure.hour, trains[i].departure.minute);
                found = 1;
```

```
        }
      }
    }
    if (!found)
      printf("No trains found in the next 1 hour.\n");
}

// Function to list all trains between a start and end station
void listTrainsBetweenStations(struct TRAIN_INFO trains[], int n, char
start[], char end[]) {
    int i, found = 0;
    printf("\nTrains between '%s' and '%s':\n", start, end);
    for (i = 0; i < n; i++) {
      if (strcmp(trains[i].start_station, start) == 0 &&
        strcmp(trains[i].end_station, end) == 0) {
        printf("%d - %s (Dep: %02d:%02d, Arr: %02d:%02d)\n",
            trains[i].train_no, trains[i].train_name,
            trains[i].departure.hour, trains[i].departure.minute,
            trains[i].arrival.hour, trains[i].arrival.minute);
        found = 1;
      }
    }
    if (!found)
      printf("No trains found between these stations.\n");
}

int main() {
    int n, i, choice;
    struct TRAIN_INFO trains[50];
    char station[50], start[50], end[50];
    struct TIME time;

    // Input number of trains
    printf("Enter number of trains: ");
    scanf("%d", &n);

    // Input train details
    for (i = 0; i < n; i++) {
      printf("\nEnter details for Train %d:\n", i + 1);
      printf("Train Number: ");
      scanf("%d", &trains[i].train_no);
      printf("Train Name: ");
      scanf(" %[^\n]", trains[i].train_name);
      printf("Start Station: ");
      scanf(" %[^\n]", trains[i].start_station);
      printf("End Station: ");
      scanf(" %[^\n]", trains[i].end_station);
      printf("Departure Time (HH MM): ");
      scanf("%d %d", &trains[i].departure.hour, &trains[i].departure.minute);
      printf("Arrival Time (HH MM): ");
```

```
        scanf("%d %d", &trains[i].arrival.hour, &trains[i].arrival.minute);
    }

    // Menu-driven operations
    do {
        printf("\n\n--- TRAIN INFORMATION MENU ---\n");
        printf("1. List all trains from a station (sorted by train number)\n");
        printf("2. List trains from a station at a particular time\n");
        printf("3. List trains from a station within next 1 hour\n");
        printf("4. List trains between two stations\n");
        printf("5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
        case 1:
            printf("Enter station name: ");
            scanf(" %[^\n]", station);
            listTrainsFromStation(trains, n, station);
            break;

        case 2:
            printf("Enter station name: ");
            scanf(" %[^\n]", station);
            printf("Enter time (HH MM): ");
            scanf("%d %d", &time.hour, &time.minute);
            listTrainsAtTime(trains, n, station, time);
            break;

        case 3:
            printf("Enter station name: ");
            scanf(" %[^\n]", station);
            printf("Enter time (HH MM): ");
            scanf("%d %d", &time.hour, &time.minute);
            listTrainsWithinOneHour(trains, n, station, time);
            break;

        case 4:
            printf("Enter start station: ");
            scanf(" %[^\n]", start);
            printf("Enter end station: ");
            scanf(" %[^\n]", end);
            listTrainsBetweenStations(trains, n, start, end);
            break;

        case 5:
            printf("Exiting program.\n");
            break;

        default:
```

```
            printf("Invalid choice! Please try again.\n");
        }

    } while (choice != 5);

    return 0;
}
```

Output:

```
Enter number of trains: 3

Enter details for Train 1:
Train Number: 101
Train Name: Express
Start Station: Delhi
End Station: Mumbai
Departure Time (HH MM): 9 30
Arrival Time (HH MM): 18 45

Enter details for Train 2:
Train Number: 102
Train Name: Shatabdi
Start Station: Delhi
End Station: Bhopal
Departure Time (HH MM): 10 00
Arrival Time (HH MM): 15 00

Enter details for Train 3:
Train Number: 103
Train Name: Rajdhani
Start Station: Delhi
End Station: Kolkata
Departure Time (HH MM): 11 00
Arrival Time (HH MM): 20 00


--- TRAIN INFORMATION MENU ---
1. List all trains from a station (sorted by train number)
2. List trains from a station at a particular time
3. List trains from a station within next 1 hour
4. List trains between two stations
5. Exit
Enter your choice: 1
Enter station name: Delhi

Trains departing from station 'Delhi':
101 - Express (09:30)
102 - Shatabdi (10:00)
103 - Rajdhani (11:00)
```

29 Write a program to swap two elements using the concept of pointers.

C using_pointer.c > main()

```c
#include <stdio.h>

void swap(int *x, int *y);
int main() {
    int i,j;
    i=10;
    j=20;
    printf("Number before swap %d %d",i,j);
    swap(&i,&j);
    printf("\n\n Number after swap %d %d",i,j);
    return 0;
}
void swap(int *x,int *y)
{
    int temp;
    temp=*x;
    *x=*y;
    *y=temp;
}
```

<u>Output:</u>

Number before swap 10 20

Number after swap 20 10